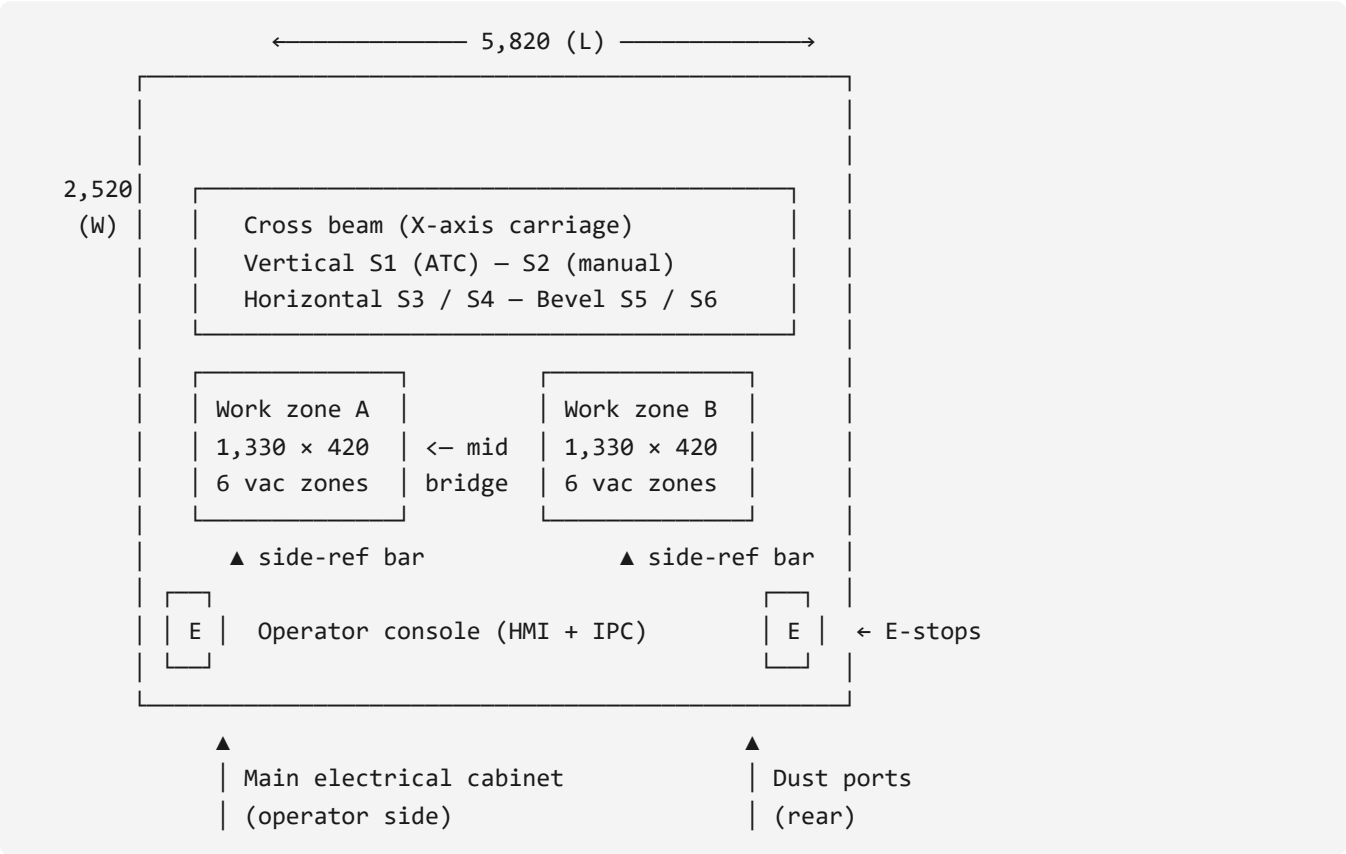


Optipeople DemoCNC D-2800M — Technical Drawings & Diagrams

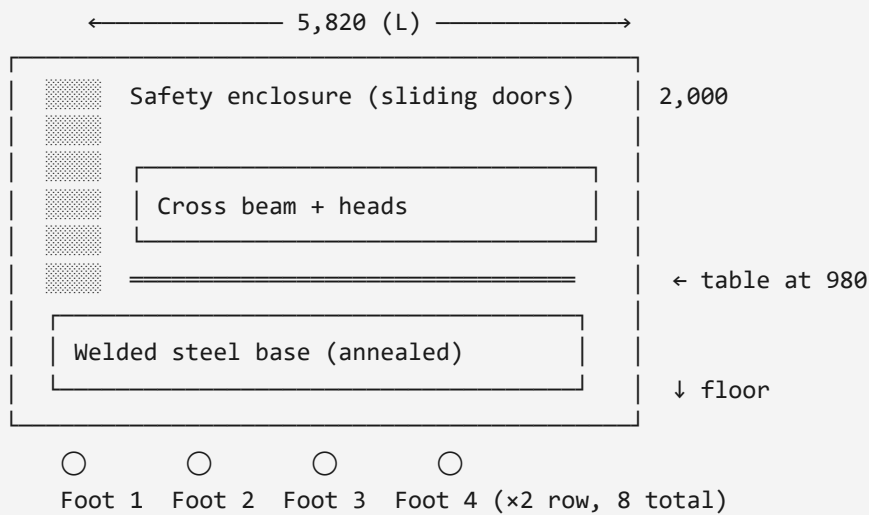
Document: D2800M-DWG-001 Revision: 3.0 Effective date: 2026-01-15

All dimensions in millimeters unless stated. Diagrams below are schematic representations. For production drawings (DXF/STEP), see drawing pack `D2800M-DWG-PACK-002` on the OptiCloud portal.

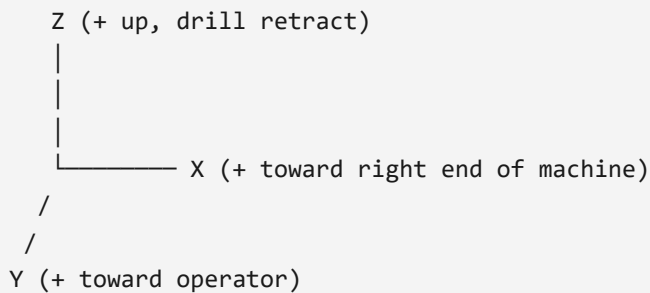
1. General arrangement — top view



2. Side elevation (front view)



3. Axis convention



- X = 0 at the home position on the left end of the cross-beam track.
- Y = 0 at the rear positioning bar.
- Z = 0 at the table top surface (auto-recalibrated by thickness sensor).
- Spindle axes (C5, C6 if installed) reported in degrees CCW.

4. Cross-beam head layout (looking at front face)

S2	S1	S3	S4	S5	S6
man.	ATC	H-mill	H-drill	bev1	bev2
vert.	vert.	6 kW	3.5 kW	2.2k	2.2k
7.5kW	7.5kW				

↑	↑	↑	↑	↑	↑
-180	-120	-60	0	+60	+120

(Y offset from S4)

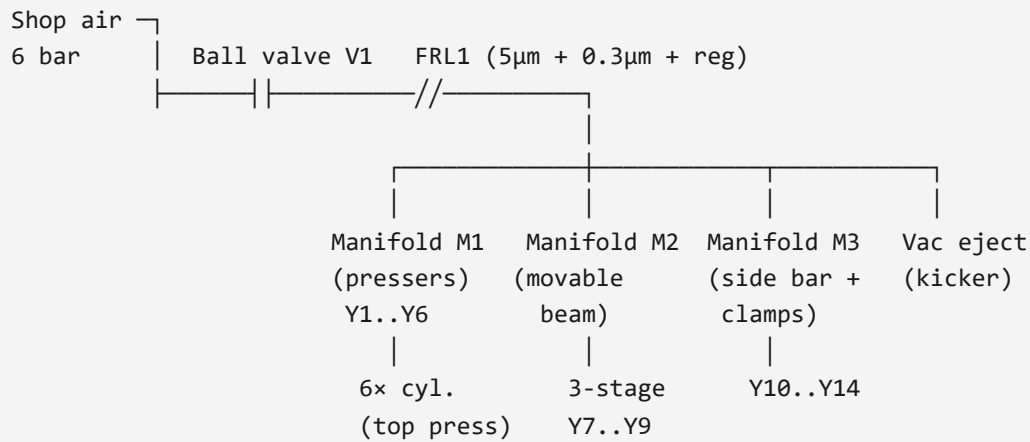
Tool magazine (8 slots, travels with the beam):

1	2	3	4	5	6	7	8
---	---	---	---	---	---	---	---

ISO30 / ER32

Spindles are mounted on a common Z-carriage and indexed in software. Offsets listed are nominal and live in `machine.cfg` → `headOffsets[]`.

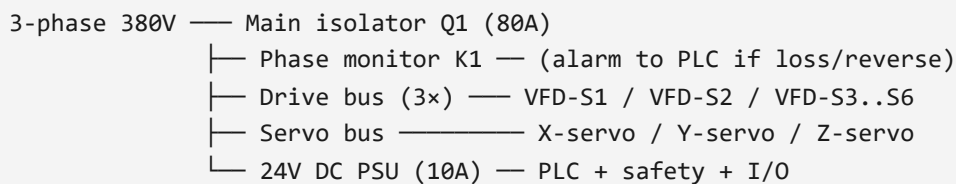
5. Pneumatic schematic (P&ID, simplified)



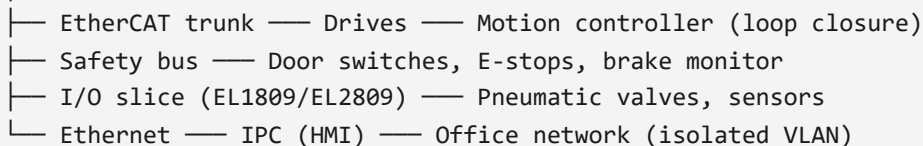
Sensors:

- PS1: input pressure, normally-open, trips < 4.5 bar
- PS2..PS7: confirm-extended on each presser cylinder
- PS8/PS9: beam-up / beam-down reed switches

6. Electrical block diagram

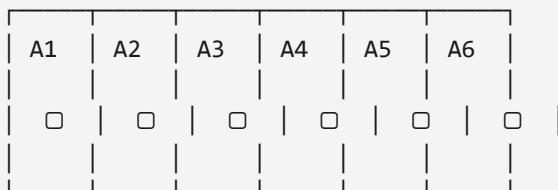


PLC (TwinCAT, FSoE master)



7. Vacuum table — zone layout

Work zone A (1330 × 420)



← V1 → ← V2 → ← V3 → ← V4 → ← V5 → ← V6 →

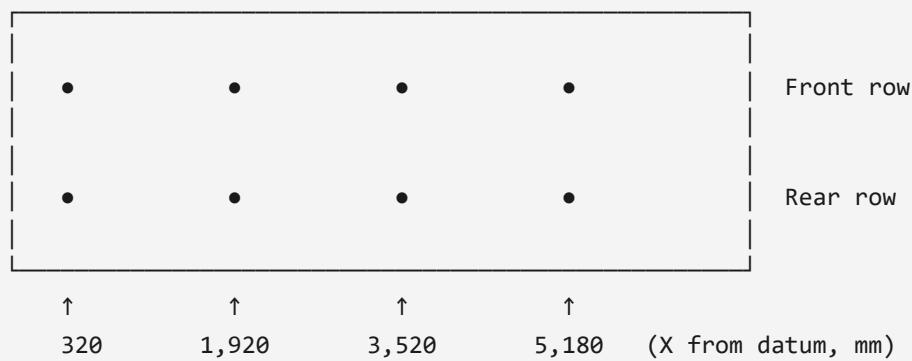
Each zone has its own 3/2 valve (V1..V6) and a non-return check.

Zones are auto selected based on workpiece outline in OptiPanel.

OptiPanel Demo CNC D-2800M — demo material

OptiPanel Demo CNC D-2800M — Technical Drawings & Diagrams

8. Foundation plan (anchor pattern)



Anchors: M16 chemical resin anchors, min embed 130 mm,
concrete C25/30 or better, min slab 200 mm thick.
Y-spacing: 1,820 mm centre-to-centre between front & rear row.

9. Hose & cable trench (recommended)

Cable trench (300 mm wide × 200 mm deep)			
Power 4×25mm ² + PE	Air pipe Ø32	Vac pipe Ø125	
Data Cat6A		Dust Ø200 (×2)	

Maintain 200 mm clearance between power and data. Vacuum and dust pipes should rise to the machine — never trap moisture in a low loop.

10. Service clearances

Side	Min clearance	Reason
Front (operator)	1,500 mm	Loading, HMI access
Rear	800 mm	Dust hoses, electrical service
Left end (X-)	1,200 mm	X-home reference, panel loading
Right end (X+)	1,200 mm	Tool magazine service
Top	500 mm above guard	Crane lift for spindle service

11. Lift points

Use the four M20 lifting eyes on the upper frame corners (red-painted). Recommended sling angle $\geq 60^\circ$ from horizontal. Centre of gravity is located ~2,400 mm from the X- end, ~1,150 mm from the rear, ~850 mm from the floor (machine on its feet). Do **not** lift by the cross beam, sheet-metal panels, or dust ducts.